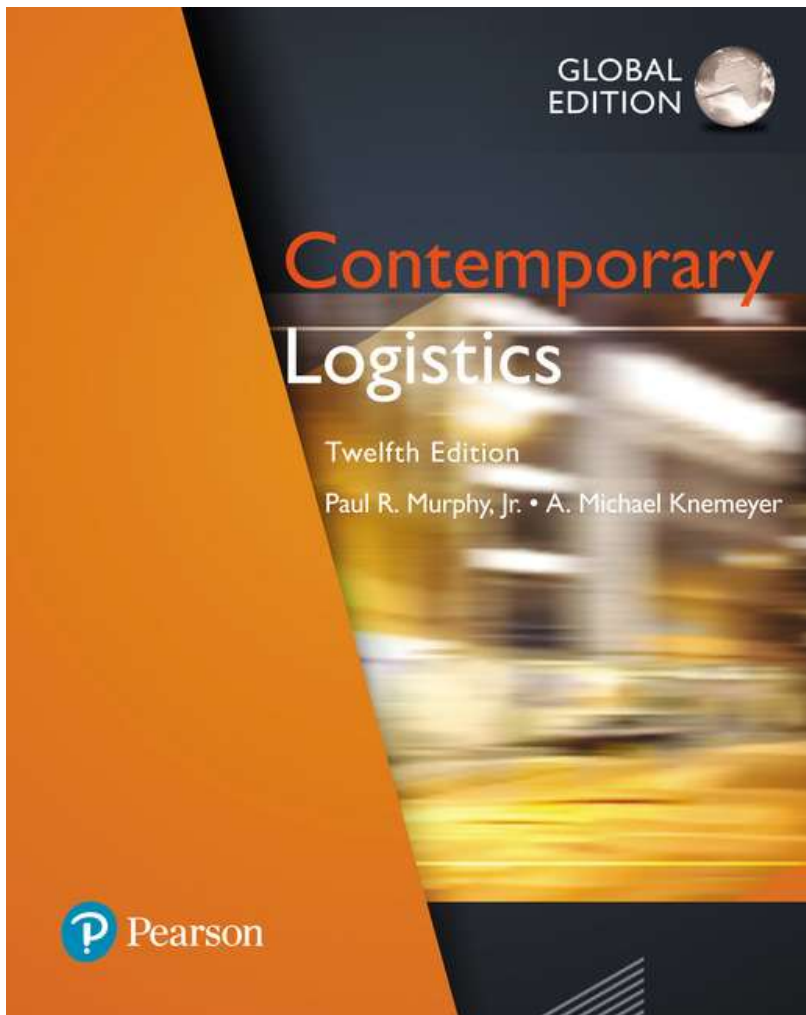


Contemporary Logistics

Twelfth Edition, Global Edition



Chapter 13

Transportation Management

Learning Objectives

- 13.1** To discuss how rates are determined
- 13.2** To learn about modal and carrier selection
- 13.3** To distinguish among various transportation documents
- 13.4** To illustrate select activities associated with making and receiving shipments
- 13.5** To learn about transportation service quality

Transportation Management (1 of 2)

- Transportation management
 - Refers to the buying and controlling of transportation service by either a shipper or consignee¹
- Transportation is the most costly logistics activity account for more than 50% of logistics cost
- Freight transportation represents 6% of GDP in USA
- Key decisions of transportation managers:
 - Mode selection
 - Carrier selection
 - Route selection
 - Rate determination

Transportation Management (2 of 2)

- Transportation managers also involved in other operations of the firm:
 - Marketing by quoting freight rates
 - Manufacturing by advising on packaging and materials handling and making certain that an adequate supply of transportation is available
 - Outbound shipping by providing simplified shipping or routing guides, drawing up transportation documents, and encouraging shipment consolidations
 - Purchasing of raw materials by advising about methods to control the costs and quality of inbound deliveries

Rate (Pricing) Considerations (1 of 10)

- Rate determination (transportation costing or pricing)
 - One key responsibility of transportation managers
 - Rate is Price charged for freight transportation
 - Fare is Prices charged for passenger transportation
- Rate determination
 - $\text{Transportation cost} = \text{Weight} \times \text{rate}$
 - Transportation rates based on three factors:
 1. Product
 2. Weight
 3. Distance

Rate (Pricing) Considerations

Approaches for rate determination (3 of 10)

1. Commodity rate

- One specific rate for every possible combination of product, weight, and distance
- Many different rates for each combination of product, weight, distance

2. Class rate system

- System to simplify rate determination
- National Motor Freight Classification (NMFC) has 18 rating classes numbered from 50 to 500 (higher rates)
- Classification numbers are used as codes understandable for carriers and shippers, and specifying packaging that must be used

M&A VISUAL GUIDE TO FREIGHT CLASSES

	Freight Class	Commodity Example	Weight Per Ft ³	
High Density ↑	50	Very Durable, Fits On 4x4 Pallet	>50lbs	Low Cost Per lb. ↓
	55	Bricks, Cement, Hard Wood	35-50lbs	
	60	Car Accessories, Printed Materials	30-35lbs	
	65	Bottled Beverages, Boxed Books	22.5-30lbs	
	70	Automobile Engines, Bulk Food	15-22.5lbs	
	77.5	Tires, Bathroom Fixtures, Thick Gloves	13.5-15lbs	
	85	Crated Machinery, Cast Iron Stoves	12-13.5lbs	
	92.5	Computers, Monitors, Refrigerators	10.5-12lbs	
	100	Vehicle Covers, Canvas, Wooden Crate	9-10.5lbs	
	110	Cabinets, Framed Artwork, Table Saw	8-9lbs	
↓ Low Density	125	Radio, Coffee Maker, Small Appliances	7-8lbs	High Cost Per lb. ↓
	150	Medical Instruments, Sheet-Metal Parts	6-7lbs	
	175	Stuffed Furniture, Clothing, Netting	5-6lbs	
	200	Aircraft Parts, Aluminum	4-5lbs	
	250	Bamboo Furniture, Mattress/Box-Springs	3-4lbs	
	300	Wood Cabinets, Chairs, Tables	2-3lbs	
	400	Paper Coffee Filters	1-2lbs	
	500	Ping Pong Balls, Styrofoam Packing	<1lb	

Figure 13.1: Page from National Motor Freight Classification

NATIONAL MOTOR FREIGHT CLASSIFICATION 100-AM		
ITEM	ARTICLE	CLASS
	GLASS: subject to item 86500	
86560	Automobile or Boat Glass , including Windshield Glass , cut to shape, see Note, item 86566, in Packages 772, 2146, 2147, 2223, 2261 or 2483:	
Sub 1	Not bent, in boxes or crates:	
Sub 2	120 united inches or less.....	70
Sub 3	Exceeding 120 united inches.....	85
Sub 4	Bent, other than tempered:	
Sub 5	Nested, in boxes, crates or Packages 304, 305, 315, 2347 or 2348	85
Sub 6	Not nested, in boxes or Packages 1180, 1247 or 2034.....	125
Sub 7	Bent, tempered, in boxes, crates or Packages 304, 305, 315, 774, 2347 or 2348, or when not nested, in Packages 1180 or 2034.....	85
86566	NOTE—Applies only on glass in specific sizes or shapes for installation in automobile bodies or in or on boats.	
86580	Building Slabs , opaque glass, NOI, with or without backing, in boxes or crates:	
Sub 1	⁵ / ₁₆ inch thick or less	85
Sub 2	Over ⁵ / ₁₆ inch thick,	65
86600	Cullet (Broken Glass) , in boxes or drums.....	50
86650	Glass , crushed, ground or powdered, not including enamel or frit, in bags, boxes, drums or Package 2453.....	55
86660	Glass , faceted, in panels, see Note, item 86662, in boxes or crates	85
86662	NOTE—Applies on glass, plain or colored, set in cement, concrete or plastic resins to form designs, patterns or pictures.	
86670	Glass , flashed or ruby, not framed nor leaded, in boxes	85
86700	Glass , flat, NOI, see Note, item 86701:	
Sub 1	Bent, in boxes or crates:	
Sub 2	220 united inches or less.....	85
Sub 3	Exceeding 220 united inches but not exceeding 15 feet in length nor 9 feet in width	100
Sub 4	Exceeding 15 feet in length or 9 feet in width.....	250
Sub 5	Not bent, see Note, item 86702, in boxes, crates or Packages 195, 198, 235, 785, 2008, 2025, 2147, 2149, 2160, 2239, 2245, 2281 or 2497:	
Sub 6	220 united inches or less.....	65
Sub 7	Exceeding 220 united inches but not exceeding 15 feet in length nor 9 feet in width	100
Sub 8	Exceeding 15 feet in length or 9 feet in width.....	200
86701	NOTE—The term 'flat' applies to glass known as sheet, plate, polished prism, rolled, window or float glass, whether or not polished, laminated, colored, opalescent, opaque, chipped, decorated, wired, etched, figured, acid dipped, ground, sandblasted, metalized (sprayed with atomized metal while glass is hot) or tempered, but not when silvered for mirrors, nor flashed, nor framed or leaded (set in or framed by lead or other metal).	
86702	NOTE—Flat glass, not bent, may also be cut to size, edges beveled or ground, or holes cut or drilled.	
86750	Glass , leaded, see Note, item 86752, in boxes:	
Sub 1	With landscape, pictorial or religious designs,	200
Sub 2	With curved, angled or straight line patterns, or with designs other than landscape, pictorial or religious	100
86752	NOTE—The term 'leaded glass' means glass either colored or clear, set in lead or in other metal.	
86770	Glass , microscope slide or slide cover, see Note, item 86771, in boxes	70
86771	NOTE—Does not apply on microscope slides or slide cover glasses. Applies only on the glass from which these articles are manufactured. For provisions applicable to microscope slides or slide cover glasses, see item 88535.	
86830	Glass , rolled, overlaid with aluminum strips with metal terminals attached, in boxes or crates	77.5
86840	Glass , rolled, overlaid with aluminum strips, NOI, in boxes or crates	70
86900	Glass , silvered for mirrors, not framed, nor backed nor equipped with hangers or fastening devices:	
Sub 1	Shock (window glass, silvered), in boxes, see Note, item 86902	85
Sub 2	Other than shock glass. in packages shown:	

Rate (Pricing) Considerations (5 of 10)

- Rate determination
 - Factors used for determine product's freight classification
 - Density
 - Refers to how heavy a product is in relation to its size
 - Viewed as primary factor for setting a product's classification
 - Lower density higher class code
 - Cubing out vs weighing out
 - Stowability
 - Refers to how easy the commodity is to pack into a load
 - Possible considerations involve the commodity's ability to be loaded with hazardous materials and ability to load freight on top of the commodity

Rate (Pricing) Considerations (6 of 10)

- Rate determination
 - Factors used for determine product's freight classification
 - Ease or difficulty of handling
 - Refers to challenges to handling that might be presented by a commodity's size, weight, and so on
 - Liability for loss and damage
 - Considers, among others, a commodity's propensity to damage other freight, its perishability, and its value



Rate (Pricing) Considerations (7 of 10)

- Rate determination
 - Weight groups are used to simplify shipment weight
 - Weight group examples:
 - Less than 500 pounds (highest rate)
 - 500–999 pounds
 - 1000–1,999 pounds
 - Distances are simplified through rate basis numbers
 - Zip codes are replacing rate basis numbers

Table 13.1: Example of the Class Rate System

Shipment	Commodity Classification	Weight (pounds)	Distance (Using ZIP Codes)	Transportation Charges ^a
1	Class 100	500	44023 to 32169	\$666.52
2	Class 200	500	44023 to 32169	\$1,245.87
3	Class 100	1,500	44023 to 32169	\$1,533.26
4	Class 100	500	44023 to 90210	\$829.96

^a Transportation charges are representative as of early 2016.

Data derived from a sample rate calculator for Old Dominion Freight Lines <http://www.odfl.com/LTLrate/LTLRateResult.faces>

Shipment 1 vs 2

Shipment 1 vs 3

Shipment 1 vs 4

3 times the weight is not 3 times more expensive.

Rate (Pricing) Considerations (8 of 10)

- Rate determination
 - Commodity Classification Standards Board develops and maintains commodity freight classifications
 - Shippers prefer lower classification number (lower rate)
 - Carriers prefer higher classification number (higher rate)
 - Transportation managers can appeal a commodity's classification

Figure 13.2: Motor Carrier Classification Docket Proposal for Changing the Classification of magnesium rakes(1 of 2)

CCSB DOCKET 2015-1

SECTION I

SUBJECT 3

Re: Rakes or Lutes, magnesium

CCSB Contact: George M. Beck Telephone — (703) 838-1813 beck@nmfta.org
Allison L. Austin (703) 838-8864 austin@nmfta.org

Proponent: Surfa Slick, LLC, Riverside, California

Present Classification Provisions

Item	Description	Class
TOOLS, OR PARTS NAMED: subject to item 183550		
Rakes , hand, having tines constructed of: subject to item 185645		
185653	Magnesium, in boxes.....	100

Proposed Classification Provisions

Item	Description	Class
TOOLS, OR PARTS NAMED: subject to item 183550		
Rakes , hand, having tines constructed of: subject to item 185645		
185653	Magnesium, etc.....	Cancel; see item NEW
NEW Rakes or Lutes , hand, consisting of magnesium heads with detached handles of same or other material; or Handles or Parts thereof , NOL; in boxes:		
Sub 1	Exceeding 96 inches in length.....	92.5
Sub 2	Not exceeding 96 inches in length	70

Analysis

This proposal was submitted by Surfa Slick, LLC to add a new item under the Tools, or Parts Named generic heading for magnesium rakes or lutes, or handles or parts thereof. The proponent's rakes or lutes are currently classified under the provisions of item 185653, naming "Rakes, hand, having tines constructed of magnesium." The proponent states, however, that these particular commodities exhibit transportation characteristics not reflected in the current class 100. The proposal seeks to establish provisions for magnesium rakes or lutes, based on the length of the handling unit.

History of Provisions

Provisions for magnesium rakes were established as a result of action taken on Docket 932, Subject 18 (March 1993). That proposal sought to establish classes for rakes based on the material construction of the tines. Information on that record showed a density of 9.47 pcf for magnesium rakes. When shipped in boxes, no unusual or significant handling, stowability or liability problems were indicated. The proposal was approved as modified, and the provisions first appeared in Supplement 2 to NMF 100-T, effective July 10, 1993. The provisions have remained substantially unchanged to the present.

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Subject 3, Page 1 of 4

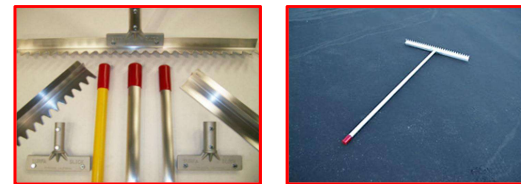
CCSB DOCKET 2015-1

SECTION I

SUBJECT 3

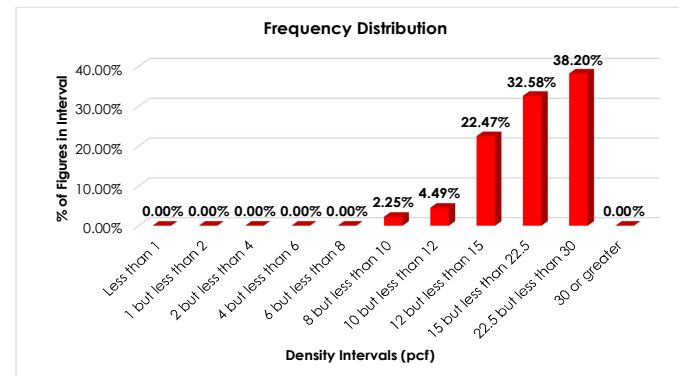
About Magnesium Rakes or Lutes

Magnesium rakes or lutes are tools primarily used in the spreading and leveling of asphalt or concrete. They may also be used to level dirt, rock, gravel or snow. The proponent's products consist of a fiberglass or magnesium handle, metal connector bracket and a magnesium rake or lute head. Some handles will exceed eight feet in length. Examples of the involved parts and assembled product are shown in the photos below.



Transportation Characteristics

Density—The proponent submitted a total of 89 density observations, including data on all of its models of magnesium rakes or lutes and the individual parts. The densities range from 9.41 to 28.46 pcf, with an average density of 18.73 pcf. A frequency distribution is shown below.



Subject 3, Page 2 of 4

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Figure 13.2: Motor Carrier Classification Docket Proposal for Changing the Classification of magnesium rakes(2 of 2)

CCSB DOCKET 2015-1

SECTION I

SUBJECT 3

When the data is evaluated based on the handling unit's length, the following density ranges and averages emerge.

Handling Unit Length	Density Range (pcf)	Average Density (pcf)
Exceeding 96 inches	9.41 – 16.57	13.39
Not exceeding 96 inches	11.63 – 28.46	20.38

Handling—As shown in the photo on the right, the proponent's magnesium rakes or lutes are shipped in fiberboard boxes secured on lift truck skids or pallets. Individual components (i.e. handles, connector brackets or heads) of magnesium rakes or lutes may be shipped in boxes on the same handling unit. Data furnished by the proponent indicates that handling units can exceed 96 inches in length. Generally, freight that exceeds 96 inches in length will require extra care and attention during handling.



Stowability—The proponent states that the involved products are stackable. However, depending on the shipment configuration, its magnesium rakes or lutes may or may not provide a regular load-bearing surface for top freight. In addition, handling units that exceed 96 inches in length may complicate the carrier's ability to structure a load and maximize vehicle utilization.

Liability—The involved articles are not perishable, hazardous in nature, unusually susceptible to theft, unusually liable to damage and would not appear likely to damage other freight. The proponent provided the list prices for various models of its magnesium rakes or lutes. The list price ranges from \$12.58 to \$13.48 per pound, with an average price per pound of \$13.16. The individual parts range in list price from \$12.00 to \$19.43 per pound, with an average price per pound of \$14.33. The proponent states that no claims have been filed for a one-year period.

Relationship to CCSB Policies and Guidelines

CCSB policy calls for classification provisions to reflect a commodity's known transportation characteristics. The proponent asserts that the provisions of item 185653 for "Rakes, hand, having tines constructed of magnesium" are not reflective of the transportation characteristics of its products. Information provided by the proponent includes 89 density observations ranging from 9.41 to 28.46 pcf, with an average density of 18.73 pcf. An average density of 18.73 pcf is generally associated with a class 70, which under CCSB guidelines calls for a minimum average density of 15 pcf. However, CCSB policies further state that unusual or significant negative handling, stowability or liability characteristics may be contributing factors in the assignment of classes. In this regard, it is noted that some of the involved magnesium rakes or lutes exceed 96 inches in length, which present additional handling and stowability considerations. In such situations, classification precedent has been to assign a class higher than that called for under the CCSB density guidelines. Surfa Slick has proposed classes accordingly, as outlined in the table on the following page.

CCSB DOCKET 2015-1

SECTION I

SUBJECT 3

Handling Unit Length	Average Density (pcf)	CCSB Minimum Average Density Guideline (pcf)	Class Based on CCSB Guidelines	Class Adjustment Based on Handling and Stowability Considerations
Exceeding 96 inches	13.39	12	85	92.5
Not exceeding 96 inches	20.38	15	70	—

This proposal would cancel item 185653 with reference to a new item for "Rakes or Lutes, hand, consisting of magnesium heads with detached handles of same or other material; or Handles or Parts thereof, NOI," at class 70 when the length does not exceed 96 inches, consistent with the respective density guideline. Handling units exceeding 96 inches in length would be assigned a class 92.5, one class higher than that called for by the respective density guideline.

This proposal is in keeping with CCSB policies and precedent.

Rate (Pricing) Considerations (9 of 10)

- Rate and service negotiations
 - Both rates and service levels may be negotiated due to economic deregulation and market situation
 - Allows transportation managers to take advantage of trade-offs between price and service

Rate and Service Items in the Carrier–Shipper Negotiation Process

Adjustment to Rates	Arbitration	Articles and Commodities Covered
Audit rights	Basis for charges	Billing procedures
Carrier equipment and drivers	Carrier insurance	Carrier notification requirements
Detention time	Duration of agreement	Estimated transportation volume
How loss and damage claims are handled	Lead times	Pallet loading
Proof of delivery	Renegotiation and reopening of contract	Schedule of rates and charges
Termination of agreement	Transportation service level	Waiver of terms



Rate (Pricing) Considerations

Rate and service negotiations (10 of 10)

- Domestic terms of sale: involves when freight charges are paid for a particular domestic shipment
 - **Free on Board (FOB) origin** buyer assumes title and control of a shipment at the point of pickup
 - **FOB destination** seller retains title and control of a shipment until it is delivered
 - **FOB origin, freight collect:** buyer pays freight charges, owns the goods in transit, and files loss and damage (L&D) claims, if needed.
 - **FOB origin, freight prepaid:** seller pays the freight charges; the buyer owns the goods in transit and files L&D claims, if needed.
 - **FOB origin, freight prepaid and charged back:** seller pays the freight charges in advance but bills the buyer for them; the buyer owns the goods in transit and files L&D claims, if needed.
 - **FOB destination, freight collect:** The buyer pays the freight charges; the seller owns the goods in transit and files L&D claims, if needed.
 - **FOB destination, freight prepaid:** The seller pays the freight charges, owns the goods in transit, and files L&D claims, if needed.
 - **FOB destination, freight prepaid and charged back:** The seller pays the freight charges in advance but bills the buyer for them; the seller owns the goods in transit and files L&D claims, if needed.

Modal and Carrier Selection (1 of 2)

- Two-step process
 1. Determine appropriate mode(s) based on modal characteristics presented in Chapter 12 would be helpful in the modal selection process.
 2. Select carrier(s) within the chosen mode(s)
- Carrier selection is more challenging
 - Difficult to be aware of every possible carrier
 - Lack of agreement on the number of relevant factors, e.g., total transit time, transit time reliability, competitive rates, and loss and damage performance.

Modal and Carrier Selection (2 of 2)

- Recently, **amodal** shipper concept started to appear:
 - Refers to a transportation manager who purchases a prespecified level of transportation service (eg. 2 days delivery for a price) and is indifferent to the mode(s) and/or carrier(s) used to provide the actual transportation service
 - Amodal concept is gaining popularity due to logistics providers such as Fedex and UPS who owns and operate different transportation modes (air, expedited, LTL, parcel)
 - Research indicates shippers are more interested in transportation metrics (e.g, transit time and its dependability) than in modes

Documentation (1 of 6)

- Documentation
 - Serves practical function
 - What, where, and how much is being transported
 - Potentially provides legal recourse
 - Transportation department is responsible for completing all documents needed to transport the firm's products
 - Carriers provide software or secure websites that enable the shipper to generate all the commonly used documents
 - Some shippers also have their own order processing software that is capable of generating transportation documents

Documentation (2 of 6)

- Bill of lading
 - Most important single transportation document
 - Functions as a delivery receipt when products are tendered to carriers
 - On freight receipt, the carrier signs the bill of lading and gives the original to the shipper
- Bills of lading types:
 - **Straight bill of lading** the name of the shipment receiver (consignee) is stated and the carrier is under strict legal obligation to deliver the freight to only the named consignee
 - **Order bill of lading** name is not specified but shipment is tendered to a carrier with a destination (used for slow-paying customer)

Documentation (3 of 6)

- Bill of lading
 - Additional classifications:
 - Long-form
 - Short
 - Preprinted

Figure 13.3: A Long-Form Bill of Lading

[illegible]

Credit: Burlington Northern Santa Fe Railroad

Documentation (4 of 6)

- Freight bill
 - Invoice submitted by the carrier requesting to be paid
 - Freight bill-paying service
 - Automated service in attempt to pay carriers within a specific number of working days

Documentation (5 of 6)

- Freight bill
 - Internal audits
 - Conducted to:
 - Detect current errors that result in overcharges
 - Correct these errors in the future
 - External audits
 - Work is performed by an independent third party to detect errors in freight bills

Documentation (6 of 6)

- Freight claims
 - Refers to a document that notifies a carrier of wrong or defective deliveries, delays, or other delivery shortcomings
 - Concealed loss or damage difficult for shippers and carriers
 - Refers to situation where loss or damage is not apparent until after a shipment has been unpacked and inspected

Making and Receiving Shipments (1 of 5)

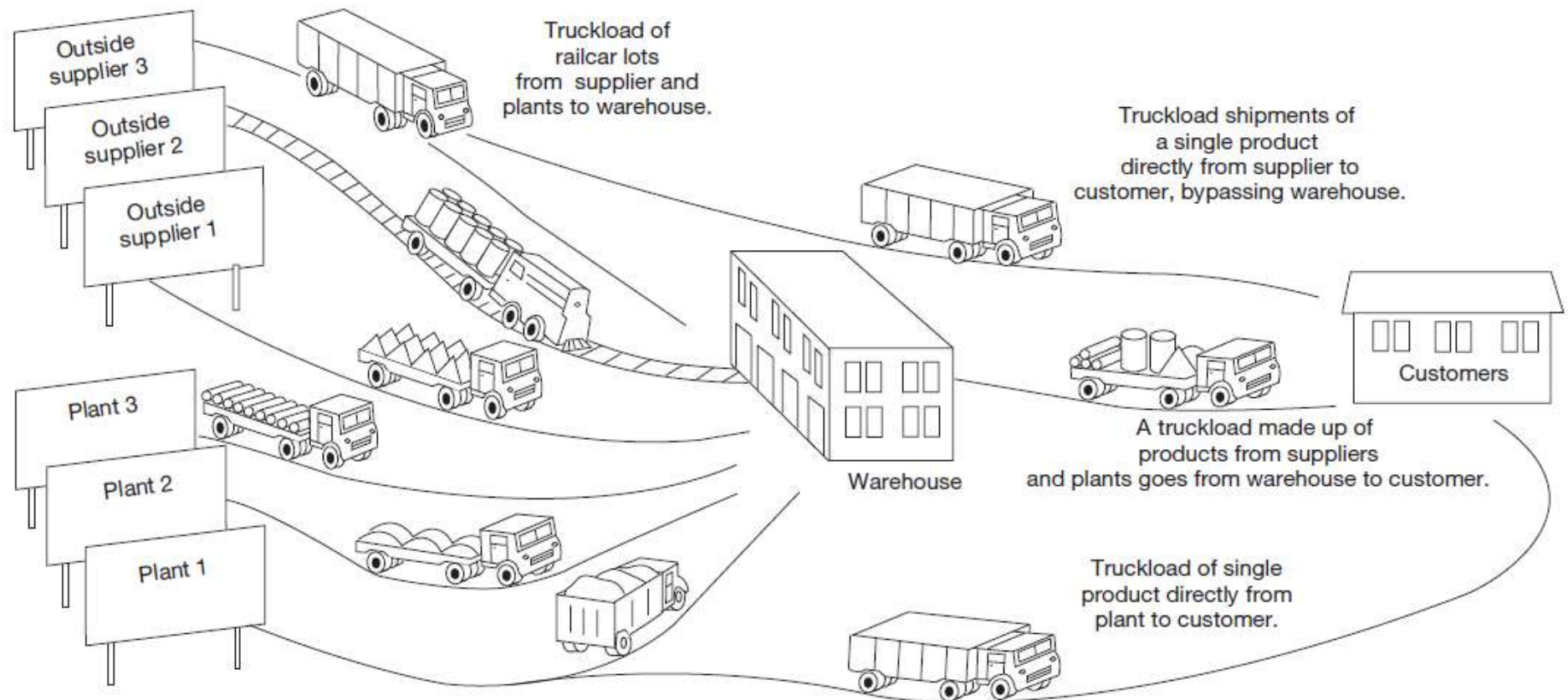
- Key area of decision making in transportation management involves making and receiving shipments
 - Refers to tactical planning and control of shipments along with supervision of freight loading and unloading²

²John E. Tyworth, Joseph L. Cavinato, and C. John Langley, *Traffic Management: Planning, Operations, and Control* (Prospect Heights, IL: Waveland Press, 1991).

Making and Receiving Shipments (2 of 5)

- Consolidating small shipments
 - Shipments more than 50 pounds and less than 500 pounds
 - To get a lower rate, shipment consolidation may occur: aggregating customer orders across time or place

Figure 13.4: The Transportation Manager Consolidates Shipments Whenever Possible



Making and Receiving Shipments (3 of 5)

- Demurrage and detention
 - Demurrage
 - Penalty payment made to the railroad for keeping a railcar beyond the time when it should be released back to the carrier
 - Detention
 - Same concept as demurrage except it usually refers to the trucking industry

Making and Receiving Shipments (4 of 5)

- Routing
 - Process of determining how a shipment will be moved between origin and destination³
 - Routing guide
 - Example of routing
 - Document that can provide a variety of shipment-related information

³http://www.cscmp.org/sites/default/files/user_uploads/resources/downloads/glossary.pdf

Making and Receiving Shipments (5 of 5)

- Tracking and expediting
 - Tracking
 - Refers to determining a shipment's location during the course of its move
 - Expediting
 - Involves the need to rapidly move a shipment to its final destination

Transportation Service Quality

- Macroenvironmental changes have caused organizations to demand higher levels of service quality
- Economic deregulation allowed for both price and service competition resulting in a need to measure their carriers' performance
- Can measure performance through the use of performance scorecards

Table 13.3: Example of a Carrier Performance Scorecard

Attribute	Importance (%)	Evaluation (1–5)	Weighted Evaluation
Improved cost performance	30	4	1.2
On-time delivery performance	25	4.5	1.125
Customer complaints	20	4.25	.85
Loss and damage performance	15	4.75	.7125
Social media presence	10	2.5	.25
Sum			4.1375
Performance score			82.75%

Key Terms (1 of 2)

- Amodal shipper
- Bill of lading
- Class rate system
- Commodity rate
- Concealed loss or damage
- Demurrage
- Density
- Detention
- Documentation
- Expediting
- FOB destination

Key Terms (2 of 2)

- FOB origin
- Freight bill
- Freight claims
- Rate
- Routing
- Routing guide
- Stowability
- Tracking
- Transportation Management